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Article Information

Drug-induced autoimmune hemolytic anemia associated with piperacillin-tazobactam: a case report and systematic review

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Abstract

Background: Piperacillin-tazobactam is a widely used broad-spectrum antibiotic. Drug-induced autoimmune hemolytic anemia (AIHA) is a rare and potentially life-threatening adverse event.

Case: 62-year-old male developed severe AIHA after 10 days of piperacillin-tazobactam for hospital-acquired pneumonia.

Presentation: Fatigue, jaundice, dark urine.
Labs: Hemoglobin 6.2 g/dL, reticulocyte count 12.5%,
DAT positive for IgG and C3d, LDH 1,250 U/L.

Treatment: Drug discontinued, corticosteroids initiated.
Hemoglobin normalized over 4 weeks.

Systematic Review: 23 previously reported cases, mortality 8.7%.

Case Presentation

Patient: Robert J. Kellerman, 62-year-old Caucasian male
Medical History: Type 2 diabetes, COPD, chronic kidney disease

Hospital Course:

- Day 1: Admitted with hospital-acquired pneumonia
- Day 1: Started on piperacillin-tazobactam 4.5g IV q6h

- Day 5: Improving respiratory symptoms
- Day 8: Developed fatigue, dark urine
- Day 10: Jaundice noted, hemoglobin dropped to 6.2 g/dL (baseline 13.8 g/dL)

Workup:

- Direct antiglobulin test (DAT): Positive for IgG and C3d
- Reticulocyte count: 12.5%
- LDH: 1,250 U/L (normal: 140-280)
- Haptoglobin: <10 mg/dL (normal: 30-200)
- Peripheral smear: Spherocytosis

Diagnosis: Drug-induced autoimmune hemolytic anemia

Management:

1. Piperacillin-tazobactam discontinued immediately
2. Prednisone 1mg/kg/day initiated
3. Transfusion of 2 units PRBC
4. Close monitoring of hemoglobin and renal function

Outcome: Hemoglobin normalized to 12.1 g/dL over 4 weeks.
Prednisone tapered over 8 weeks. No recurrence.

Discussion

Drug-induced AIHA is rare (estimated 1 per 100,000 drug exposures). Piperacillin-tazobactam is the second most commonly implicated antibiotic after ceftriaxone. Mechanism involves drug adsorption onto red blood cell membrane, triggering immune response.

Mortality rate in published cases: 8.7% (2/23).
All fatalities occurred in patients with comorbidities.
Early recognition and drug discontinuation are critical.